

Division Operation

$$Q(Y) = R(Z) \div S(X)$$

Where $X \subset Z$ and $Y = Z - X$

For a subtuple $t[Y]$ in R

to appear in the result Q , the

$t[Y]$ must appear in R in

combination with every tuple $t[X]$

in S .

ex)

WORKS-ON'

ESSN	PNO
------	-----

PROJECT'

PNO

Q: List the SS# of employees
who work on all the projects.

Q

ESSN

$Q = \text{WORKS-ON}' \div \text{PROJECT}'$

ex)

R	A	B
	a	10
	a	15
	b	7
	c	10
	c	15

S	B
	10
	15

$$R(A, B) \div S(B) = \begin{array}{c|c} Q & A \\ \hline & a \\ & c \end{array}$$

Since $Q(Y) = R(Z) \div S(X)$

$$= \pi_Y(R) - \pi_Y((\pi_Y(R) \times S) - R),$$

$$Q(A) = R(A, B) \div S(B) = \pi_A(R) - \pi_A((\pi_A(R) \times S) - R)$$

$$\pi_A(R) = \begin{array}{c|c} & A \\ \hline & a \\ & b \\ & c \end{array}$$

$$\pi_A(R) \times S = \begin{array}{c|c} & A & B \\ \hline & a & 10 \\ & a & 15 \\ & b & 10 \\ & b & 15 \\ & c & 10 \\ & c & 15 \end{array}$$

$$(\pi_A(R) \times S) - R = \begin{array}{c|c} & A & B \\ \hline & b & 10 \\ & b & 15 \end{array}$$

$$\pi_A((\pi_A(R) \times S) - R) = \begin{array}{c|c} & A \\ \hline & b \end{array}$$

$$\pi_A(R) - \pi_A((\pi_A(R) \times S) - R) = \begin{array}{c|c} & A \\ \hline & a \\ & b \\ & c \end{array} - \begin{array}{c|c} & A \\ \hline & b \end{array}$$

$$= \begin{array}{c|c} & A \\ \hline & a \\ & c \end{array}$$

Relational Algebra Question 8.16

Q 8.16

$$a) \quad R_1 \leftarrow \sigma_{PNAME = 'productX'} (PROJECT)$$

$$R_2 \leftarrow \pi_{ESSN} \left(\sigma_{HOURS > 10} (WORKS_ON \bowtie R_1) \right)$$

$PNO = PNUMBER$

$$R_3 \leftarrow \sigma_{DNO = 5} (EMPLOYEE)$$

$$R \leftarrow \pi_{FNAME, LNAME} (R_2 \bowtie R_3)$$

$ESSN = SSN$

b)

$$R \leftarrow \pi_{FNAME, LNAME} (EMPLOYEE \bowtie DEPENDENT)$$

$SSN = ESSN$ AND
 $FNAME = DEPENDENT_NAME$

$$c) \quad R_1 \leftarrow \pi_{SSN} \left(\sigma_{FNAME = 'Franklin' \text{ AND } LNAME = 'Wong'} (EMPLOYEE) \right)$$

$$R \leftarrow \pi_{FNAME, LNAME} (EMPLOYEE \bowtie R_1)$$

$SUPERSSN = SSN$

d)

$$R (PNAME, SUM_HOURS) \leftarrow$$

$$PNAME \neq SUM_HOURS (WORKS_ON \bowtie PROJECT)$$

$PNO = PNUMBER$

$$e) \quad R_1 (PNO) \leftarrow \pi_{PNUMBER} (PROJECT)$$

$$R_2 \leftarrow \pi_{ESSN, PNO} (WORKS-ON) \div R_1$$

$$R \leftarrow \pi_{FNAME, LNAME} (EMPLOYEE \bowtie_{SSN=ESSN} R_2)$$

$$f) \quad R_1 \leftarrow \pi_{ESSN} (WORKS-ON)$$

$$R_2 \leftarrow \pi_{SSN} (EMPLOYEE)$$

$$R_3 (ESSN) \leftarrow R_2 - R_1$$

$$R \leftarrow \pi_{FNAME, LNAME} (EMPLOYEE \bowtie_{SSN=ESSN} R_3)$$

$$g) \quad R \leftarrow \pi_{DNAME} \gamma_{AVERAGE\ SALARY} (EMPLOYEE \bowtie_{DNO=DNUMBER} DEPARTMENT)$$

$$h) \quad R \leftarrow \gamma_{AVERAGE\ SALARY} (\sigma_{SEX='F'} (EMPLOYEE))$$

$$i) \quad R_1 \leftarrow \pi_{ESSN} (WORKS-ON \bowtie_{PNO=PNUMBER} (\sigma_{PLocation='Houston'} (PROJECT)))$$

$$R_2 \leftarrow \pi_{DNUMBER} (DEPARTMENT) - \pi_{DNUMBER} (\sigma_{DLOCATION='Houston'} (DEPT-LOCATIONS))$$

$$R \leftarrow \pi_{FNAME, LNAME, ADDRESS} (R_1 \bowtie_{ESSN=SSN} EMPLOYEE \bowtie_{DNO=DNUMBER} R_2)$$

$$j) R_1 (MGRSSN) \leftarrow \pi_{MGRSSN} (DEPARTMENT) - \pi_{ESSN} (DEPENDENT)$$

$$R \leftarrow \pi_{LNAME} (EMPLOYEE \bowtie_{SSN = MGRSSN} R_1)$$

k. Generalize Query i above to list the names and addresses of employees who work on a project in one city but whose department has no location in that city.

$$k) R_1 \leftarrow \pi_{ESSN, LOCATION} (WORKSON \bowtie_{PNO = PNUMBER} PROJECT)$$

$$R_2 \leftarrow \pi_{SSN, LOCATION} (EMPLOYEE \bowtie_{DNO = DNUMBER} DEPT-LOCATIONS)$$

$$R_3 (ESSN, LOC) \leftarrow R_1 - R_2$$

$$R \leftarrow \pi_{FNAME, LNAME, Address} (EMP \bowtie_{SSN = ESSN} R_3)$$